Team Details

Team Name: PIXELS

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	Tatikole Ashwini	2nd Year
2	Member 1	Vanisha Nadimpalli	2nd Year
3	Member 2	Yadagiri Navadeep Saran	2nd Year
4	Member 3	Ravi Tanuja	2nd Year

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Problem Statement Addressed

Edge-AI based defect classification system for semi conductor wafer/die images

•**Defect Types**: Semiconductor wafers contain multiple defect classes—shorts, opens, bridges, vias, scratches, cracks, and LER that impact product quality.
•**Current Limitation**: Manual visual inspection by human operators is time-consuming, inconsistent, and prone to human error.
•**Quality Impact**: Missed defects during inspection lead to faulty products reaching customers, causing warranty claims and brand reputation damage.
•**Industry Criticality**: Semiconductors are essential for electronics, IoT, automotive, and AI applications—quality failures have cascading effects.
•**Production Bottleneck**: Manual inspection creates delays in manufacturing workflow, reducing throughput and increasing operational costs.
•**Need for Automation**: Requires an automated, AI-powered defect detection system delivering high accuracy with real-time processing on manufacturing floors.

Idea Description -

Edge-optimized AI model automatically classifies semiconductor wafer defects in real-time.

KEY CONCEPT & APPROACH

Developing a lightweight, transfer learning-based CNN model that automatically classifies semiconductor wafer defects into multiple categories and is optimized for real-time edge deployment via ONNX format with minimal computational overhead.

SOLUTION OVERVIEW

An edge-optimized AI defect classification system that utilizes transfer learning (MobileNetV2/EfficientNet-Lite) to automatically detect and categorize semiconductor wafer defects in real-time (<100ms inference, <10MB model size), replacing slow manual inspection with accurate, scalable automated quality control directly on manufacturing equipment.

Proposed Solution

Deep learning CNN model for automated semiconductor wafer defect detection, optimized for edge deployment with real-time inference.

We propose an Edge-AI–based defect classification system using lightweight deep learning to automatically identify wafer defect patterns from semiconductor inspection images. The solution is optimized for fast, low-memory inference suitable for edge deployment in manufacturing environments.

Dataset Plan:
Total images planned/current: 27900
No. of classes: 8
Class list: Bridges + CMP_Scratches+cracks+LER+Malformed_Vias+Opens + Clean + Other
Class balance plan: Balanced class weights (inverse frequency) to handle natural imbalance ratio of up to 18,865:24(786:1)
Train/Val/Test split: 80/10/10
Image type: Semiconductor microscopy images (SEM/AFM)
Labelling method/source: CLIP-based zero-shot classification + automated organization

Technology & Feasibility/Methodology Used

A lightweight MobileNetV2 model is trained using transfer learning on a public semiconductor wafer dataset and evaluated using accuracy, precision, recall, and a confusion matrix on the test set.

Model details:

Architecture: MobileNetV2+Custom classification Head
Training approach: Transfer learning + class-Weighted loss
Input size: 224 x 224 x 3
Model size: 9.11 MB
Framework: Training: TensorFlow/Keras
Deployment: ONNX (Edge AI - NXP eIQ compatible)

Metrics on your test split:

Accuracy: 69.93%
Precision/Recall: 85%/70%
Confusion Matrix: (image)
https://drive.google.com/file/d/1iZ8y4ObF5GhkJQMyXZHAayQpOO3evA_P/view?usp=drive_link

Artifacts & Links

GitHub Repository

https://github.com/Vanisha-18/semiconductor-defect-classification-edge-ai

Prototype / Simulation Video

https://drive.google.com/file/d/1qByS3AvmUSoQKtkTevO7nTg5aLYHllcC/view?usp=sharing

Dataset ZIP link

https://drive.google.com/drive/folders/1uKFTmw25lH3mjKaPWMHHpKNHDYhizfzs?usp=drive_link

ONNX model link

https://drive.google.com/file/d/1o3DDbjcMov8xbpt6PkzYStKMcLOTmJl3/view?usp=drive_link

Results report link

https://docs.google.com/document/d/1GwZR1t4I2iKUrBogeYhUHgmXlFUSo5haxoOUYyzSgCo/edit?usp=sharing

Research and References

Research & References

1. https://researchdata.ntu.edu.sg/dataset.xhtml?persistentId=doi:10.21979/N9/WBLTFI

2. https://github.com/wenbihan/MIIC-IAD?tab=readme-ov-file